



UNIVERSITY OF LEEDS

Formal Language & Finite Automata

Closure properties

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All reading is from the "Introduction to the theory of computation" -
Micheal Sipser

- Closure properties for regular languages - Pages 58 - 63 (essential)

Closure properties

The concept of closure is common in computer science and mathematics and appear naturally when programming.

A set A is said to be **closed** with respect to some operation if when you apply the operation to elements of the set A the result is an element of set A .

Examples

- The set of natural numbers are closed with respect to the $+$ operation.
- The set of rational numbers are closed with respect to $+$, $-$ and $\times$

Closure of regular languages

Regular languages are closed with respect to many operations.

Definition

Let A and B be two regular languages over Σ , we define the following operations,

- **Union:** $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
- **Intersection:** $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$
- **Complement:** $\bar{A} = \{x \in \Sigma^* \mid x \notin A\}$
- **Concatenation:** $A \circ B = \{xy \mid x \in A \text{ and } y \in B\}$
- **Kleene star:**
 $A^* = \{x_1 \dots x_k \mid k \geq 0 \text{ and, for each } 0 \leq i \leq k, x_i \in A\}$

Closure of regular languages

Theorem (Closure with respect to Union)

Regular languages are closed with respect to Union.

Theorem (Closure with respect to Intersection)

Regular languages are closed with respect to Intersection.

Theorem (Closure with respect to Complement)

Regular languages are closed with respect to Complement.

Closure of regular languages cont.

Theorem (Closure with respect to Concatenation)

Regular languages are closed with respect to Concatenation.

Theorem (Closure with respect to Kleene star)

Regular languages are closed with respect to Kleene star.